Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm/ Final Exam

Course Title	Control System Lab	Course Code	
Semester	First Semester	Academic Year	
Exam Date		Day: Monday	Time: 14:30-15:20

Student Name: _____ Student ID No. _____ Section _____

GENERAL INSTRUCTIONS

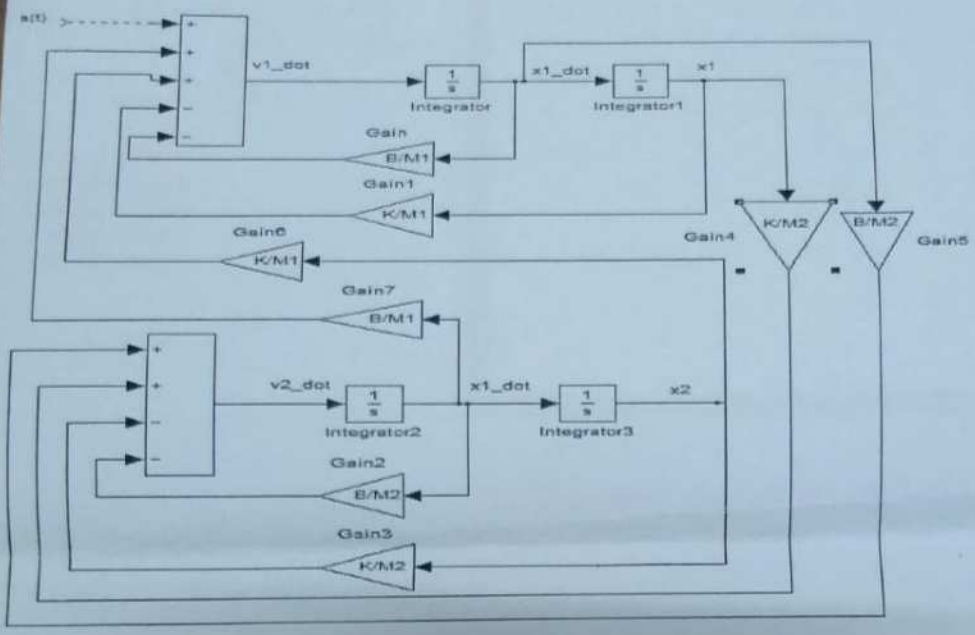
1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Essay	CLIO,1,6	10(33.3%)			
2	Essay	CLIO, 1,2,3,5	10(33.3%)			
3	Essay	CILO 1,4	10(33.4%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

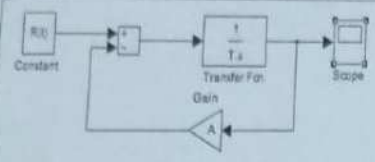
Q1 [10 marks] Write the differential equations for the following system



V1_dot =

V2_dot =

Q2) [10 marks] Plot the output response for each of the following conditions

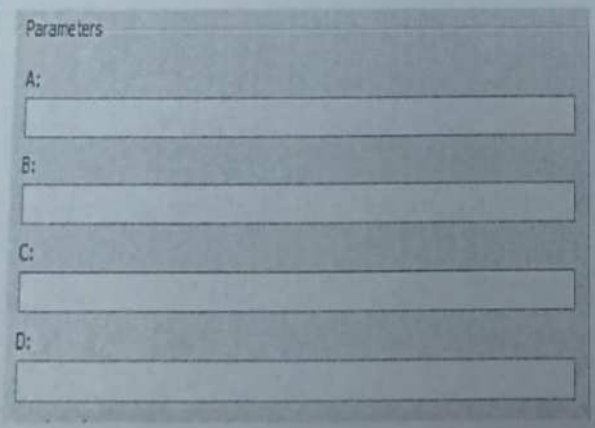


1. R(t)=1, T=1, A=0
2. R(t)=1, T=1, A=1
3. R(t)=1, T=1, A=3
4. R(t)=1, T=2, A=1
5. R(t)=2, T=1, A=1

Q 3) [10 marks] For a system given where Y(t) is system output and m(t) is system input ;

$$y'''(t) - 2y'(t) + 5y(t) - 3m(t) = 0$$

a. Determine the state space matrices



b. Solve this equation by transfer function